

Waals Forces pull them together. And chemical bonding only allows atoms to be arranged in certain ways. None of these forces are noticeable at the macroscopic level, but they will restrict the way we can build with atoms.

## Grainy Matter

You cannot always extrapolate common ideas, devices, or principles down to the nano-scale. Physical laws behave differently, so what works up here may not down there. Take a simple mechanical device such as a bearing. A common ball bearing consists of a ring of metal balls that slide around a circular track. To rotate smoothly, the touching surfaces must be very smooth. If we build an atomic counterpart made of round atoms, we cannot “smooth” the mating surfaces. The metal balls and the track are individual atoms, and you cannot subdivide an atom. Matter is “grainy” at the atomic level. In a real-world bearing, graphite or oil is used as a lubricant. In an atomic bearing, the graphite atoms would be as large as the bearing; no lubrication is possible. In short, we cannot assume that real world tools can be reduced to nano-scale. The physical laws are different.

## Knowledge

According to some, what will stop nanotechnology is not physics but the limits of human comprehension. Nature is almost infinitely complex. And nanotechnology hopes to upstage nature; to intentionally design everything about an object that takes nature thousands of years to create using random experimentation (evolution). Consider a simple natural object: an apple. To “nano build” an apple, you would need an atomic map of an apple containing trillions of parts. You would then have to design proteins that could learn that map and place each atom perfectly. Understanding such a complex system may be beyond the limits of human comprehension.

If you doubt there are such limits, consider this. Humans have tried to plan a less complex, natural object: an economy. The Soviet-style central planning system was a massive failure by any standard, despite massive amounts of information and complete power. Economists in the West have had only limited success at just predicting economic conditions. Nanotechnology is a far more ambitious goal with far more information to be managed.

One might think that anything scientific has rules and can be learned eventually, but it is not so. No scientist can predict exactly how a cloud will move. The Navier-Stokes Equations, a mathematical approximation of how fluids like clouds should move, cannot be solved. The ultimate design limit on nanotechnology may be that some things just cannot be done because they are too complex. 